

VENKATA KRISHNA CHAITANYA KUMAR PADARTHI

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Summary

As an Industrial Engineering graduate with a solid foundation in statistical analysis and data visualization, I have successfully transitioned from a 2-year tenure as a Java Developer to a 1-year role as a Graduate Research Assistant. My fully funded project at Universal Fiber Systems involved developing the Allstream Digital Foundation and Root-Cause Analysis for Scrap Reduction, leveraging my expertise in Python and data science concepts. This initiative resulted in a predictive model for manufacturing machine failure, accessible through a Python web application. Specializing in manufacturing and supply chain processes, my commitment to data-driven excellence positions me as a valuable asset, ready to contribute to the growth and success of any organization through continuous process improvements and operational efficiency.

EDUCATION

Master's in industrial engineering – 3.83/4.0 GPA

Aug 2022 – May 2024

J.B. Speed School of Engineering, University of Louisville, Kentucky, USA

Courses: Operations Research, Predictive Analytics-I, Predictive Analytics-II, Experimental Design, DSS with Excel, Probability, Data Mining with Linear Models, Modelling, Validation & Prediction, Data Visualization, Advance Manufacturing Systems, Supply Chain.

Research Advisor: Dr. Xiaoyu Chen

Email: xchen325@buffalo.edu

Bachelor's in mechanical engineering – 71.56%

Aug 2015 – Jun 2019

M.V.S.R Engineering College, Osmania University, Hyderabad, India

Research Advisor: Dr. Saparey Gajanana

Email: saparey72@gmail.com

Dissertation: 'Development of Regression Model for Al6061 +SiC Tool and Optimization Process Parameters on EDM'.

SKILLS

Programming Languages	Java, Python, JavaScript, HTML, JSP, CSS, ZUL, VBA, AJAX, R, MATLAB
Web Development	JSP, Flask, ZK, Bootstrap
Framework	Spring, Hibernate, Spring Boot
Database Management	PostgreSQL, MySQL, SQLite, PostgreSQL
Version Control, Server & Deployment	SVN, Tomcat, RedHat, MySQL

Key Competencies: Data Analysis, Machine Learning, Data Visualization, Artificial Intelligence, Data Science, Documentation & Presentation, Data Engineering, Database Management, VBA, Web Development, Version Control

EXPERIENCE

Graduate Student Assistant – University of Louisville, Louisville, KY, USA

Sep 2023 – Oct 2023

Alumni Data Base of Industrial Engineering Department

- Data on industrial engineering graduates was efficiently collected and maintained via social media channels, resulting in increased alumni interaction using automation process using python.
- Created an easy-to-use communication platform for department and alumni to improve networking and information exchange. Implemented data optimization strategies, team coordination, and provided frequent reports to the department.

Graduate Research Assistant – University of Louisville, Louisville, KY, USA

Aug 2022 – Jul 2023

Allstream Digital Foundation and Root-Cause Analysis for Scrap Reduction

- Designed and developed a realistic mimic database leveraging manufacturing company data, collaborated with experts to represent industry-specific requirements, developed a Python web application for data extraction and implemented functionality for data collection, consolidation, and visualization.
- Implemented initiatives to enhance root-cause identification, establishing a centralized data pool to align upstream and downstream data. Developed a robust root-cause analysis methodology, aiming to reduce scrap in one-step production lines by **1%~1.5%**, potentially increasing the success rate by approximately **55%~60%**.
- Developed a Python-based automation system for data extraction, consolidation, and visualization by connecting to multiple databases, facilitating efficient data management and analysis in a manufacturing laboratory environment.

System Engineer – Tata Consultancy Services, Chennai, TN, India

Oct 2020 – Aug 2022

- Spearheaded the development of a Java full-stack web application, TNIGRS, for the Tamil Nadu State Government, achieving a 20% improvement in system efficiency. Collaborated with clients, implemented change requests, and addressed defects to ensure seamless application operation.
- Optimized application performance by reducing database hits for specific tables through query modifications and the implementation of caching techniques, resulting in a 15% reduction in response time. Successfully upgraded the Log4J framework from version 1.x to version 2.x in both Portal and BackOffice servers.
- Played a pivotal role as a Deployer in the development team for two months, overseeing server management and deployment activities. Upgraded the development server from jdk1.7 to jdk1.8, enhancing overall system robustness and reducing deployment time by 25%.

- Among the various modules I developed, some notable ones include Land Registration, MIS, and Schedulers. I also completed change requests related to Public Grievances and the Chit Fund Module in the BackOffice, as well as Online Registration, Marriage Correction, and the Chit Fund Module in the Portal. To ensure code quality, I utilized SonarQube to inspect and maintain high standards in the codebase.

Research Assistant

May 2019 – Sep 2020

- Collaborated with Dr. S. Gajanana, Professor of Department in Mechanical Engineering, on various research projects within the field of Manufacturing and Data Science.
- Conducted extensive experimentation, involving tool preparation, manufacturing processes, data collection and analysis.
- Developed and fine-tuned statistical models to optimize process parameters, factor analysis, and data visualizations.
- Published multiple research papers in reputed journals, contributing significantly to the field of AI Manufacturing.
- Attended national and international conferences to present research findings and network with peers.

Intern – Pennar Industries Limited, Hyderabad, TG, India

Feb 2018

- Engaged in commercial and productivity improvement studies within the Business Excellence Department, Conducted comprehensive analyses and evaluations to identify areas of improvement in various manufacturing processes and operations.
- Utilized data-driven approaches and statistical techniques to assess the company's performance, identify sag-rod threading, Laser cutting machine and recommend effective strategies for enhancing commercial outcomes and productivity.
- Collaborated with cross-functional teams and stakeholders to gather insights, understand challenges, and develop innovative solutions to drive operational efficiency and maximize profitability.

Intern – Asaco Private Limited

Aug 2017

- Proficient in using SolidWorks for accurate modeling of machine components and assemblies.
- Skilled in operating and managing a diverse range of machines in the machine shop, including lathe, milling, drilling, and boring machines.
- Experienced in fabrication, assembly, and CNC machining, ensuring high-quality and precise manufacturing processes.

ACADAMIC PROJECTS

Master's Course Project

- Project on the topic 'Predicting Manufacturing Machine Failure using Statistical and Machine Learning Methods'. Utilized statistical and machine learning techniques to develop a predictive model for manufacturing machine failure, improving production efficiency and minimizing downtime by analyzing sensor data and implemented machine learning algorithms to accurately forecast machine failures, enabling proactive maintenance and reducing production losses.
- Project on the topic 'Testing Aluminum Silicon Carbide to Explore Alternate Electrode Materials in the Electrical Discharge Manufacturing Process'. Proficient in Electrical Discharge Manufacturing (EDM) process, utilizing Minitab for precision machining of hard materials, optimizing parameters, and reducing surface roughness in the aerospace industry. Experienced in electrode material selection and researching cost-effective alternatives like Aluminum Silicon Carbide composites.
- Project on the topic 'The Diet Problem'. Proficient in using linear programming techniques to optimize and find the most cost-effective combination of food items to meet daily nutritional requirements. Skilled in creating a user-friendly interface and streamlining the user experience through Excel VBA for efficient solving of diet problems.
- Project on the topic 'Best Chip Expansion Strategy'. Executed an Operations Research course project on developing the Best Chip expansion strategy, applying optimization techniques to maximize operational efficiency, minimize costs, and drive strategic growth for the company.
- Project on the topic 'Investigating the Impact of Nano Powder Additives in Blended Dielectric for Al6061+SiC Tool on Electrical Discharge Machining Performance.' The study focused on evaluating the influence of incorporating nano powder additives in a blended dielectric for Al6061+SiC tool on the performance of electrical discharge machining (EDM). Through rigorous experimentation and statistical analysis, including the application of linear regression using Minitab software, the impact of these additives on key EDM parameters, such as material removal rate, surface roughness, and electrode wear rate, was quantitatively assessed.

Bachelor's Project

May 2019

- Project on the topic 'Development of Regression Model for Al6061 +SiC Tool and Optimization Process Parameters on Electro Discharge Machining'. The Regression Model enables to reduce the number of experiments to achieve the required minimal surface roughness, maximum metal removal rate using different tools. Which helps to reduce the cost, time, labor and many more.

RESEARCH PUBLICATIONS

- "Investigation of Al 2024 Metal Spinning on Surface Roughness using Conical Mandrel." Manufacturing Technology Today, Vol 19, No 11 (2020).
- "Investigation and Analysis of Al6061 Tool with Blended Dielectric Fluid on EDM" Volume 11, Issue 12 (2020) Compliance Engineering Journal.
- "Regression Modelling and Numerical Simulation of Metal Spinning Process on Al2024 Alloy" Volume 11, Issue 11 (2020) Compliance Engineering Journal.
- "Comparative Study of Aluminum Alloy 6061 Tool with Graphite, Copper and Brass Tools in EDM" (September 2020) IJMPERD.
- "Effect of Nano Powder Additives in Blended Dielectric for Al6061+ Sic Tool on EDM." (2020) Studies in Indian Place Names (UGC Care Journal).
- "Development Of Regression Model for Al6061+ Sic Tool And Optimization Of Process Parameters On Electro Discharge Machining" IOP Publishing (2020).
- "Investigation on effects of process parameters on surface roughness for spinning process using cylindrical mandrel." (October 2019) MTT Journal.
- "Regression model for surface roughness on EDM using Al6061 + SiC, Copper and Brass Tools." (September 2019) MTT Journal.

CONFERENCES

- International Conference on Science Technology and Management-(ICSTM-20).
- 2nd Indian Materials Conclave Conference Organized by Material Research Society of India (MRSI-2020).
- International Conference on Recent Trends in Sciences, Engineering and Mathematics, GOA, INDIA (ICRTSEM-2019).